

LAB MANUAL

Performance Analysis of Type-1 and Type-2 Hypervisors

Proxmox VE (Type-1) vs VMware Workstation (Type-2)

This lab manual guides the learner through creating identically configured virtual machines on a Type-1 hypervisor (Proxmox VE) and a Type-2 hypervisor (VMware Workstation), and then measuring and comparing their CPU performance using Sysbench.

PART A: PERFORMANCE ANALYSIS USING TYPE-1 HYPERVISOR – PROXMOX VE

1. Accessing the Proxmox VE Web Interface

Proxmox VE is deployed on a centralized physical server and can be accessed remotely through its web-based management interface.

Procedure

1. Connect the system to the network where the Proxmox VE server is accessible.
2. Open a supported web browser.
3. Enter the following URL in the address bar:

```
https://<PROXMOX_SERVER_IP>:8006
```

Example:

```
https://192.168.X.X:8006
```

4. Press Enter.

Note

The Proxmox VE server IP address and login credentials must be provided before beginning the experiment.

2. Accessing the Proxmox VE Login Page

A browser security warning may appear because Proxmox VE uses a self-signed SSL certificate.

Click Advanced.

Select Proceed to <Server IP>.

The Proxmox VE login page will be displayed.

3. Logging in to Proxmox VE

Enter the assigned login credentials.

Parameter	Value
Username	Provided Credentials
Password	Provided Credentials
Realm	As Configured

Click Login.

After successful authentication, the Proxmox VE management dashboard will be displayed.

4. Understanding the Proxmox VE Interface

The left-side navigation panel contains the Proxmox VE infrastructure hierarchy.

```
Datacenter
|
+-- Proxmox Node
    |
    +-- Virtual Machines
    +-- Storage
    +-- Network
```

The Proxmox VE interface provides access to:

- Datacenter resources
- Proxmox server node
- Virtual machines
- Storage resources
- Network configuration
- CPU and memory utilization

5. Creating a Virtual Machine in Proxmox VE

A virtual machine is created using the Create VM wizard available in the Proxmox VE web interface.

The VM creation process consists of the following stages:

```
General -> OS -> System -> Disks -> CPU -> Memory -> Network -> Confirm
```

Step 1: Select the Proxmox Node

From the left-side navigation panel:

1. Expand Datacenter.
2. Select the required Proxmox server node.

Example:

```
Datacenter -> pve
```

The node summary page will be displayed.

Step 2: Open the Create VM Wizard

Click the Create VM button located in the upper-right corner of the Proxmox VE interface.

The Create: Virtual Machine configuration wizard will open.

6. Configuring General Settings

The first section of the VM creation wizard is General.

Configure the following parameters:

Parameter	Configuration
Node	Selected Proxmox Node

Parameter	Configuration
VM ID	Automatically Assigned / Allocated ID
Name	Unique Virtual Machine Name

Recommended Naming Convention

<Name>-Type1

Example:

CC-Experiment1-Type1

After completing the configuration, click Next.

7. Configuring the Operating System

The next section is OS.

Procedure

1. Select Use CD/DVD Disc Image File (ISO).
2. Select the storage location containing the ISO image.
3. Select the required Ubuntu ISO image.

Example:

Storage: local
ISO Image: ubuntu-22.04.iso

The configuration should be:

Parameter	Configuration
Installation Media	ISO Image
Storage	local
Operating System	Ubuntu

Click Next.

8. Configuring System Settings

The System section configures the virtual machine hardware platform.

Use the default settings unless otherwise specified.

Parameter	Configuration
Graphics Card	Default
Machine	Default
BIOS	Default
SCSI Controller	Default

Click Next.

9. Configuring Virtual Disk

The Disks section is used to configure virtual storage for the VM.

Configure the following parameters:

Parameter	Configuration
Storage	local-lvm / Assigned Storage
Disk Size	20 GB
Bus/Device	Default

Click Next.

10. Configuring CPU Resources

The CPU section is used to allocate virtual processor resources.

Configure:

Parameter	Value
Sockets	1
Cores	2

Therefore: Total vCPU = 2

Click Next.

11. Configuring Memory Resources

The Memory section is used to allocate RAM to the virtual machine.

Configure:

Parameter	Value
Memory	2048 MiB

Equivalent memory allocation: 2048 MiB = 2 GB RAM

Click Next.

12. Configuring Network

The Network section configures the virtual network interface.

Configure:

Parameter	Value
Bridge	vmbr0
Model	Default / VirtIO

The vmbr0 bridge connects the virtual machine to the configured network.

Click Next.

13. Confirming Virtual Machine Configuration

The final section is Confirm.

Review the complete configuration.

Resource	Configuration
VM Name	CC-Experiment1-Type1
Operating System	Ubuntu
CPU	2 vCPU
Memory	2 GB
Disk	20 GB
Network	vmbr0

Verify the configuration and click Finish.

The virtual machine will be created.

14. Verifying the Created Virtual Machine

After successful VM creation:

1. Locate the newly created virtual machine in the left-side navigation panel.
2. Select the virtual machine.

The VM status and configuration details will be displayed.

Example hierarchy:

```
Datacenter
|
+-- pve
    |
    +-- local
    +-- local-lvm
    |
    +-- VM-ID (CC-Experiment1-Type1)
```

15. Starting the Virtual Machine

Select the created virtual machine.

Click Start from the top-right corner.

The VM status will change from Stopped to Running.

16. Opening the Virtual Machine Console

After starting the VM:

1. Select the virtual machine from the navigation panel.
2. Select Console from the available VM options.

The virtual machine display will open within the browser.

The Ubuntu installation interface will be displayed.

17. Installing Ubuntu Operating System

Complete the Ubuntu installation using the following general procedure:

1. Select the required language.
2. Select Install Ubuntu.
3. Configure the keyboard layout.
4. Select the required installation type.
5. Select the virtual disk for installation.
6. Configure the timezone.
7. Create the operating system user account.
8. Complete the installation process.
9. Restart the virtual machine.

After restarting, log in to the Ubuntu virtual machine.

18. Verifying the Virtual Machine

Open the terminal in the Ubuntu virtual machine and execute:

```
hostnamectl
```

Verify the following information:

- Hostname
- Operating System
- Kernel Version
- Architecture

19. Analyzing CPU Configuration

Execute:

```
lscpu
```

Observe the following parameters:

- Architecture
- CPU(s)
- CPU model
- Virtualization information

Record the output for performance analysis.

20. Analyzing Memory Configuration

Execute:

```
free -h
```

Observe:

- Total memory
- Used memory
- Free memory
- Available memory

Record the output.

21. Analyzing Disk Configuration

Execute:

```
df -h
```

Observe:

- Filesystem
- Total disk capacity
- Used disk space
- Available disk space

Record the output.

22. Monitoring System Resource Utilization

Execute:

```
top
```

Observe:

- CPU utilization
- Memory utilization
- Running processes
- Load average

Press:

```
q
```

to exit the monitoring interface.

23. Installing Sysbench

Sysbench is used to perform CPU performance analysis.

Update the package repository:

```
sudo apt update
```

Install Sysbench:

```
sudo apt install sysbench -y
```

Verify the installation:

```
sysbench --version
```

24. Performing CPU Performance Analysis

Execute the following benchmark:

```
sysbench cpu --cpu-max-prime=20000 run
```

Record the following values:

- Total execution time
- Total number of events
- Events per second
- Latency statistics

25. Observation Table – Type-1 Hypervisor

Parameter	Observation
Hypervisor	Proxmox VE
Hypervisor Type	Type-1
Guest Operating System	Ubuntu
CPU Allocation	2 vCPU
Memory Allocation	2 GB
Disk Allocation	20 GB
Total Execution Time	
Total Events	
Events per Second	
Average Latency	

26. Monitoring VM Resources from Proxmox VE

Return to the Proxmox VE web interface.

Navigate using:

```
Datacenter -> Proxmox Node -> Virtual Machine -> Summary
```

Observe the following resource utilization parameters:

- CPU Usage
- Memory Usage
- Network Traffic
- Disk Usage

Record the observations for comparison with the Type-2 hypervisor.

27. Shutting Down the Virtual Machine

After completing the analysis, shut down the virtual machine properly.

Execute the following command inside the Ubuntu VM:

```
sudo poweroff
```

Alternatively, use the Shutdown option available in the Proxmox VE interface.

Verify that the VM status changes to Stopped.

PART B: PERFORMANCE ANALYSIS USING TYPE-2 HYPERVISOR – VMWARE WORKSTATION

VMware Workstation is a Type-2 hypervisor that runs on top of a host operating system.

In this experiment, a virtual machine is created using VMware Workstation with the same configuration used for the Proxmox VE virtual machine.

The performance of the virtual machine is then analyzed using Sysbench.

1. Launching VMware Workstation

Open VMware Workstation from the installed applications.

The VMware Workstation home interface will be displayed.

Select: Create a New Virtual Machine

The New Virtual Machine Wizard will open.

2. Selecting the Virtual Machine Configuration

The wizard displays the following options:

- Typical (recommended)
- Custom (advanced)

Select: Typical (recommended)

Click: Next

3. Selecting the Guest Operating System Installation Method

The next window displays the installation media options.

Select: Installer disc image file (iso)

Click: Browse

Navigate to the location where the Ubuntu ISO file is stored and select the required Ubuntu ISO image.

Example:

ubuntu-22.04.iso

After selecting the ISO file, click: Next

4. Selecting the Guest Operating System

If VMware detects the operating system automatically, verify the detected configuration.

Otherwise, select:

Parameter	Configuration
Guest Operating System	Linux
Version	Ubuntu 64-bit

Click: Next

5. Naming the Virtual Machine

Enter a name for the virtual machine.

Recommended naming convention:

CC-Experiment1-Type2

Select the location where the virtual machine files will be stored.

Click: Next

6. Configuring Virtual Disk Capacity

The disk configuration window will be displayed.

Configure the following:

Parameter	Configuration
Maximum Disk Size	20 GB
Disk Storage	Store virtual disk as a single file / default option

Click: Next

The virtual machine configuration summary will be displayed.

7. Customizing Virtual Machine Hardware

Before completing the VM creation process, click:

Customize Hardware

The Virtual Machine Settings window will open.

The following hardware resources must be configured.

7.1 Configuring Memory

From the left-side panel, select: Memory

Set the memory allocation to:

2048 MB

Equivalent to: 2 GB RAM

7.2 Configuring Processor Resources

From the left-side panel, select: Processors

Configure:

Parameter	Value
Number of Processors	1

Parameter	Value
Number of Cores per Processor	2

Therefore: Total Virtual CPUs = 2

7.3 Verifying the Virtual Disk

From the left-side panel, select: Hard Disk

Verify that the configured disk size is: 20 GB

7.4 Configuring Network Adapter

Select: Network Adapter

For this experiment, select the required network configuration.

Recommended configuration:

NAT

NAT allows the virtual machine to access the internet through the host system.

The network configuration may also be changed according to the laboratory environment.

7.5 Verifying Hardware Configuration

Verify the following configuration before closing the hardware settings window.

Resource	Configuration
Memory	2 GB
CPU	2 vCPU
Hard Disk	20 GB
Network	NAT
Guest Operating System	Ubuntu

Click: Close

The New Virtual Machine Wizard will be displayed again.

8. Completing Virtual Machine Creation

Review the virtual machine configuration.

Click: Finish

The newly created virtual machine will appear in the VMware Workstation library.

9. Starting the Virtual Machine

Select the created virtual machine from the VMware Workstation library.

Click: Power on this virtual machine

The Ubuntu installation process will begin.

10. Installing Ubuntu Operating System

The Ubuntu installation interface will be displayed inside the VMware virtual machine.

Complete the installation using the following procedure.

Step 1: Select Language

Select the required language.

Click: Install Ubuntu

Step 2: Configure Keyboard Layout

Select the appropriate keyboard layout.

Click: Continue

Step 3: Select Installation Type

Select the appropriate installation option.

For a standard installation, select:

Normal Installation

Click: Continue

Step 4: Configure Installation Disk

Select: Erase disk and install Ubuntu

Note: This operation affects only the virtual hard disk created for the VMware virtual machine.

Click: Install Now

Confirm the disk changes when prompted.

Step 5: Select Time Zone

Select the appropriate geographical location and timezone.

Click: Continue

Step 6: Create User Account

Configure the Ubuntu user account.

Enter:

Parameter	Example
Name	User Name
Computer Name	cc-type2-vm
Username	User Name
Password	Secure Password

Click: Continue

Ubuntu installation will begin.

11. Restarting the Virtual Machine

After the installation is complete, select:

Restart Now

The virtual machine will restart.

Log in using the username and password created during the installation process.

12. Verifying the Virtual Machine Configuration

After logging in to Ubuntu, open the Terminal.

Execute:

```
hostnamectl
```

Verify the following information:

- Hostname
- Operating System
- Kernel Version
- Architecture

13. Verifying CPU Configuration

Execute:

```
lscpu
```

Verify the CPU configuration.

Observe the following parameters:

- Architecture
- CPU(s)
- CPU model
- Number of cores
- Virtualization type

Verify that the VM is configured with approximately 2 Virtual CPUs.

14. Verifying Memory Configuration

Execute:

```
free -h
```

Observe:

- Total memory
- Used memory
- Free memory
- Available memory

Verify that the allocated memory is approximately 2 GB.

15. Verifying Disk Configuration

Execute:

```
df -h
```

Observe:

- Filesystem
- Total disk capacity
- Used disk space
- Available disk space

Verify the virtual disk configuration.

16. Monitoring System Resource Utilization

Execute:

```
top
```

Observe the following system parameters:

- CPU utilization
- Memory utilization
- Running processes
- Load average

Press:

```
q
```

to exit.

17. Installing Sysbench

Sysbench is used to perform CPU performance analysis.

Update the package repository:

```
sudo apt update
```

Install Sysbench:

```
sudo apt install sysbench -y
```

Verify the installation:

```
sysbench --version
```

The installed Sysbench version will be displayed.

18. Performing CPU Performance Analysis

Execute the following CPU benchmark:

```
sysbench cpu --cpu-max-prime=20000 run
```

Allow the benchmark to complete.

The output will contain CPU performance statistics.

Record the following values:

- Total execution time
- Total number of events
- Events per second
- Minimum latency
- Average latency
- Maximum latency

19. Observation Table – Type-2 Hypervisor

Record the benchmark results in the following table.

Parameter	Observation
Hypervisor	VMware Workstation
Hypervisor Type	Type-2
Guest Operating System	Ubuntu
CPU Allocation	2 vCPU
Memory Allocation	2 GB
Disk Allocation	20 GB
Total Execution Time	
Total Events	
Events per Second	
Average Latency	

20. Monitoring Resource Utilization in VMware Workstation

Return to the VMware Workstation interface.

Select the running virtual machine.

The VM status and resource allocation can be observed through:

```
VM -> Settings
```

Verify the configured hardware resources:

- Processors
- Memory
- Hard Disk
- Network Adapter

For additional observation, resource utilization may also be monitored from the Ubuntu virtual machine using:

```
top
```

or:

```
free -h
```

21. Recording the Type-2 Hypervisor Results

Record the final benchmark results obtained from the VMware Workstation virtual machine.

Type-2 Hypervisor Performance Results

Performance Metric	Result
Hypervisor	VMware Workstation
Hypervisor Type	Type-2
CPU Configuration	2 vCPU
Memory Configuration	2 GB
Disk Configuration	20 GB
Total Execution Time	
Total Events	
Events per Second	
Minimum Latency	
Average Latency	
Maximum Latency	

22. Shutting Down the Virtual Machine

After completing the performance analysis, shut down the virtual machine properly.

Execute:

```
sudo poweroff
```

Wait until the virtual machine shuts down completely.

Alternatively, use the VMware Workstation power options:

```
VM -> Power -> Shut Down Guest
```

Type-2 Hypervisor Workflow

The complete VMware Workstation procedure can be summarized as follows:

```
Launch VMware Workstation
|
Create a New Virtual Machine
|
Select Typical Configuration
|
Select Ubuntu ISO
|
Configure VM Name and Location
|
Configure 20 GB Virtual Disk
|
Customize Hardware
|
Configure 2 vCPU
|
Configure 2 GB RAM
|
Configure Network
|
Finish VM Creation
|
Power On Virtual Machine
|
Install Ubuntu
|
Verify CPU, Memory and Disk
|
Install Sysbench
|
Run CPU Benchmark
|
Record Performance Results
|
Shut Down Virtual Machine
```